

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

Mathematical
ols

Basic Math and Calculus

LECTURE-1

PHYSICS

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Basic Math
Calculus



Topics To Covered

- 1** Baat cheet on ist chapter
- 2** fraction
- 3** power
- 4** angle

The lines of life

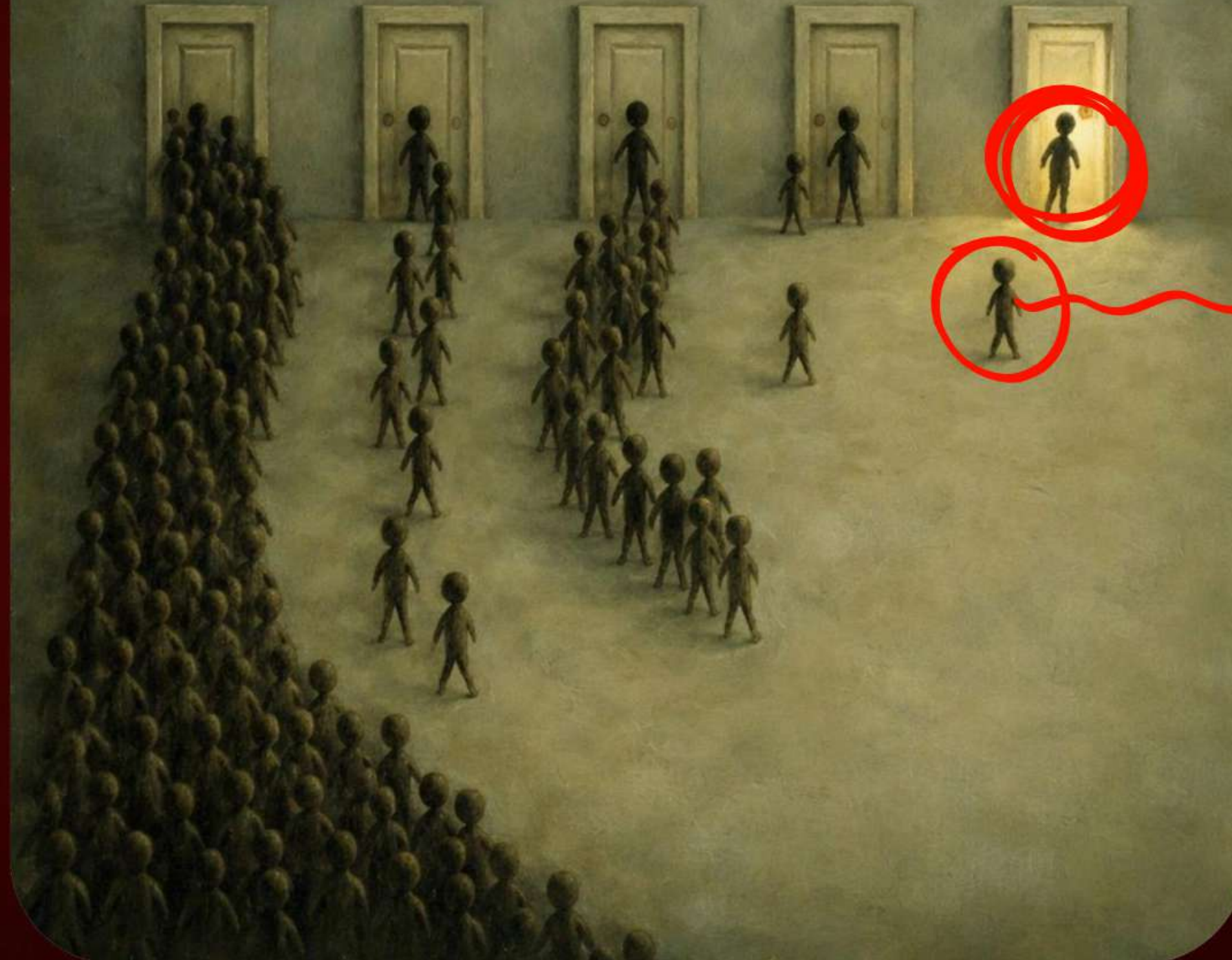
To criticize

To gossip

To get involved

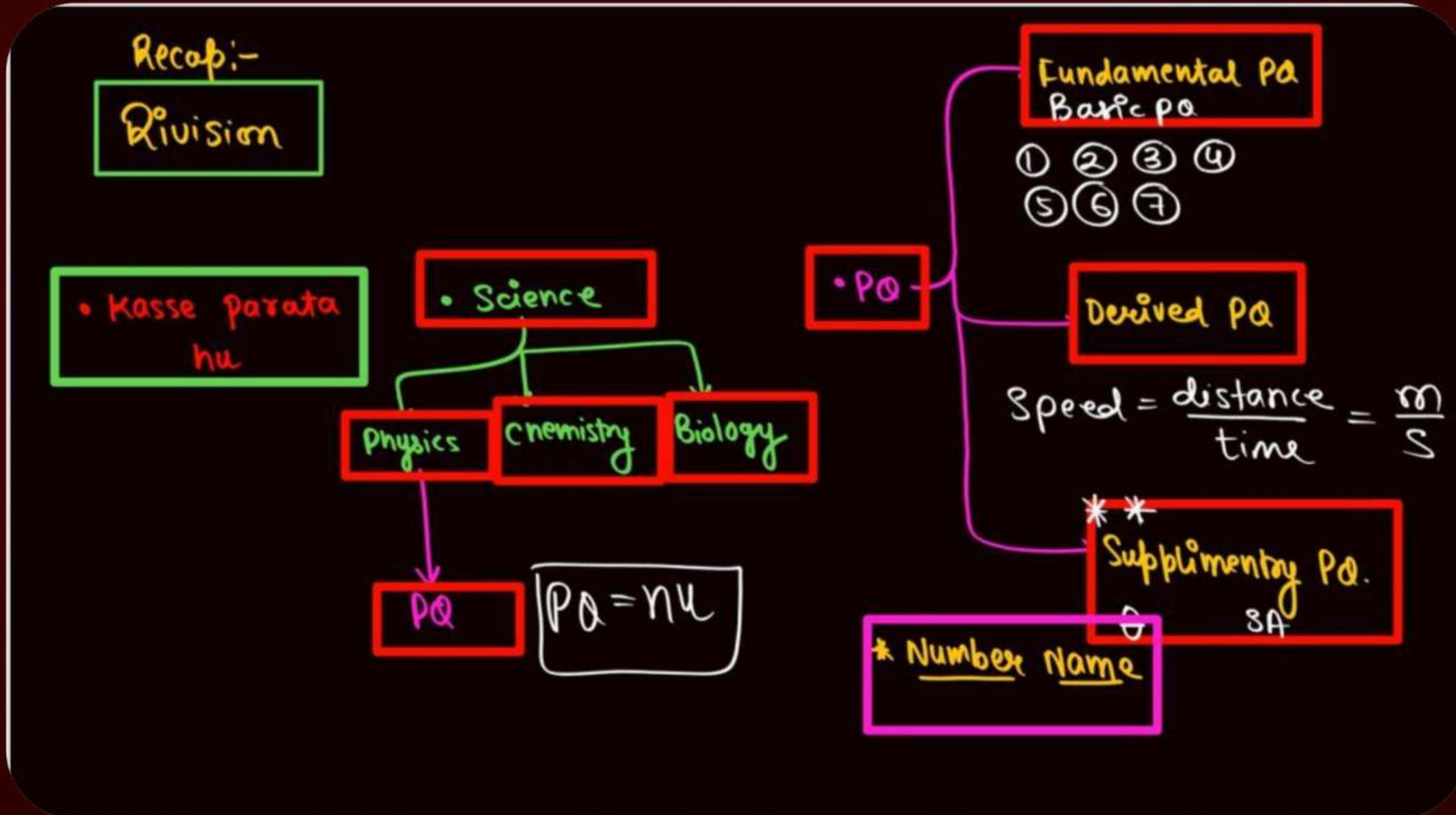
To encourage

To help

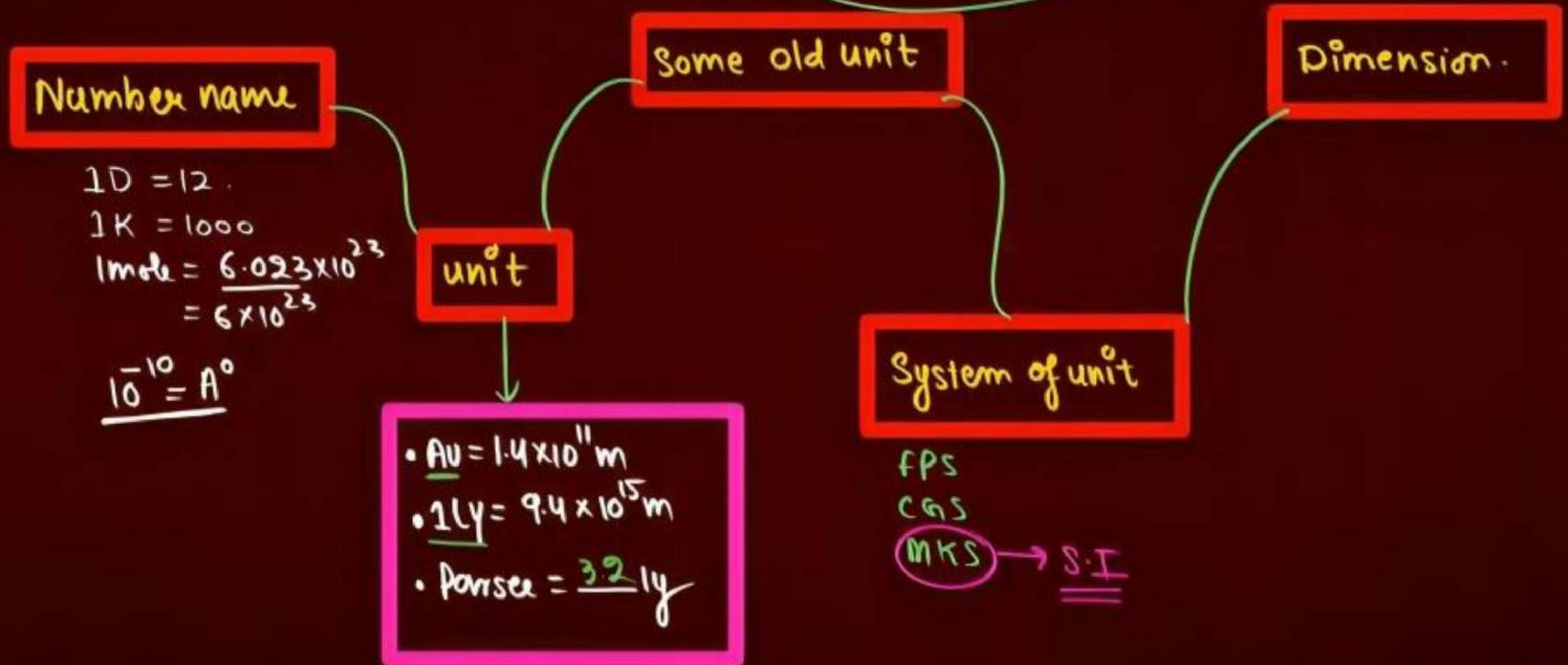


wajahid
Mastere

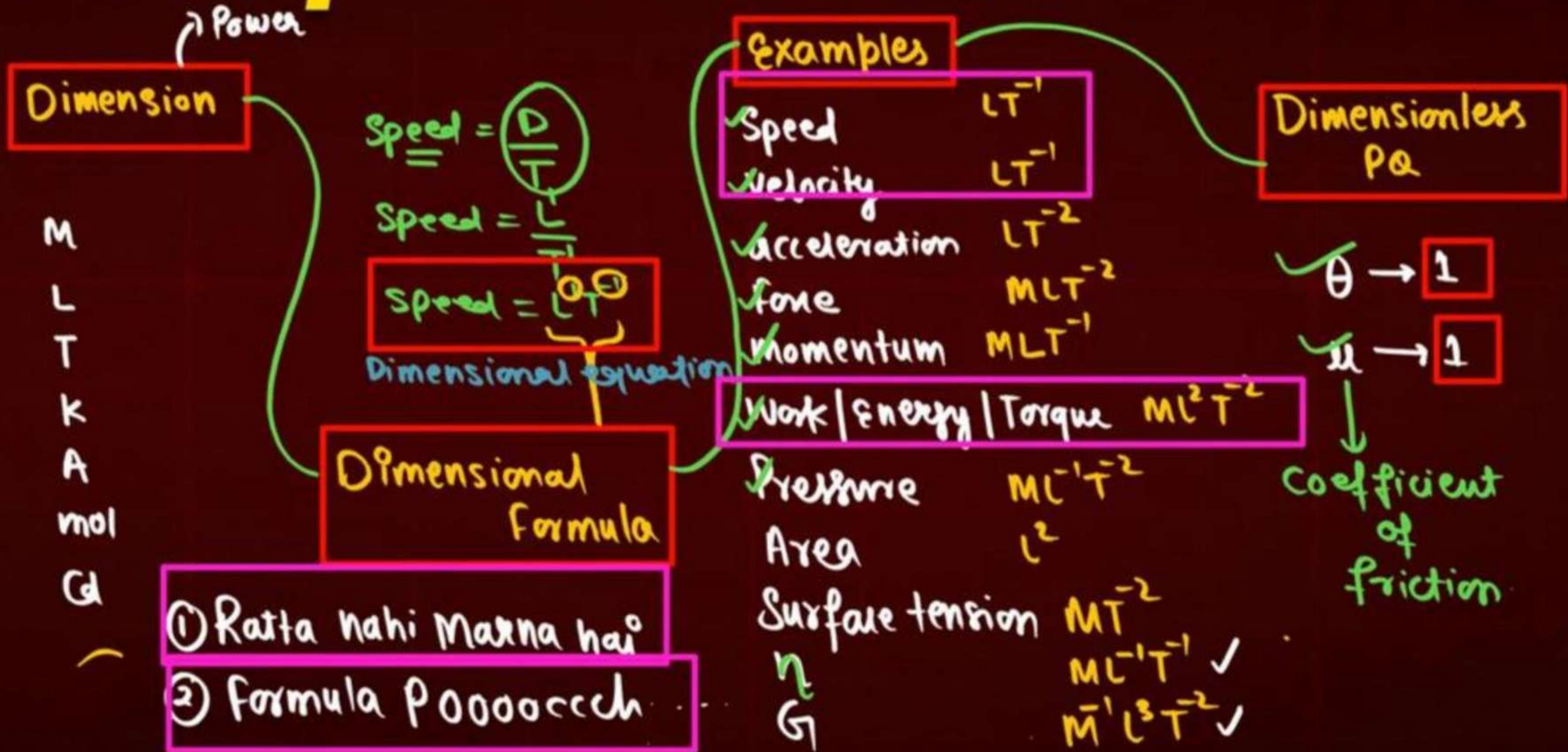
Recap From 1st Lec.



Recap From 2. Lec.



Recap From 3rd Lec.



Recap From 4th Lec.

Application
of
Dimension

a ✓

b ✓

c

d

a To find out the
Dimensional
Formula

$$\eta \rightarrow \text{ML}^{-1}\text{T}^{-1}$$

$$G \rightarrow \text{ML}^3\text{T}^{-2}$$

b To check the
Correctness of
Physics Formula

Law of
Homogeneity

$$x + x = 2x$$

$$2x - x = x$$

① $v = u + at$

$$\downarrow \quad \downarrow \quad \downarrow$$

$$\text{LT}^{-1} \quad \text{LT}^{-1} \quad \text{LT}^{-1}$$

② $s = ut + \frac{1}{2}at^2$

$$\downarrow \quad \downarrow \quad \downarrow$$

$$\text{L} \quad \text{LT}^{-1} \quad \text{LT}^{-1} \cdot \text{T}^2$$

③ $v^2 = u^2 + 2as$

$$\downarrow \quad \downarrow \quad \downarrow$$

$$\text{L}^2\text{T}^{-2} \quad \text{L}^2\text{T}^{-2} \quad \text{LT}^{-2} \cdot \text{L}$$

④ $F = \frac{mv^2}{r}$

$$\downarrow \quad \downarrow \quad \downarrow$$

$$\text{MLT}^{-2} \quad \text{MLT}^{-2} \quad \text{MLT}^{-2}$$

Recap From 5th Lec.

Application of
Dimension

© To derive the
physics formula

Time period of Pendulum

• let
 $T \rightarrow m^a l^b g^c$
 $M^0 L^0 T^1 \rightarrow M^a L^b (LT^{-2})^c$
 $a=0 \quad b=\frac{1}{2} \quad c=-\frac{1}{2}$



$2\pi = 4\pi \sqrt{l/g}$
 $T = 2\pi \sqrt{l/g}$

Stroke law

• let
 $F \rightarrow \eta^a r^b v^c$
 $MLT^{-2} \rightarrow (ML^{-1}T^{-1})^a L^b (LT^{-1})^c$
 $a=1 \quad b=1 \quad c=1$

$F = 6\pi \eta r v$

Recap From 6th Lec.

④

conversion of
one unit system
into another
unit system

$$* 1\text{m} = 100\text{cm}$$

$$1\text{kg} = 1000\text{g}$$

$$1\text{N} = 10^5 \text{ dynes (PVA)}$$

$$1\text{joule} = 10^7 \text{ erg (PVA)}$$

$$n_2 = n_1 \left[\frac{M_1}{M_2} \right]^a \left[\frac{L_1}{L_2} \right]^b \left[\frac{T_1}{T_2} \right]^c$$

* Limitation of Dimension

* we cannot derive the
Dimensional const.

* we cannot derive the
formula having $(+/-)$

* Cannot derive the formula
with $\sin\theta / \cos\theta$

* It is not unique

Recap From 7th Lec.

Rounding off

Rule ① 2.4829
 2.483

Rule ② 2.4562
 2.456

Rule ③ 5.234215 5.234225
 5.23422 5.23422

Significant Fig

Number	SF
24321	5
24000	2
34051	5
4.600	4
0.009	1
3×10^{-6}	1
number	∞ (infinite)

Recap From 8th Lec.

Scientific Notation

* Ek tarika hai to
reduce the number 10^n

$$245620. = 2.4562 \times 10^5$$

$$0.00052 = 5.2 \times 10^{-4}$$

Why Measurement is important

- ✓ To understand the science
- ✓ Building Bridge & machines
- ✓ Cooking & Shopping
- ✓ Share clear information

- Maths Bohat AhSaam hai

“**Maths** is the language of **universe**”

Maths is a **Queen**, Physics is the **King**

.

Maths

Numbers: - 0, 1, 2, 3, 4

$$\begin{array}{r} 54 \\ + 25 \\ \hline 79 \end{array} \quad \textcircled{1}$$
$$\begin{array}{r} 56 \\ + 26 \\ \hline 82 \end{array}$$

$$\begin{array}{r} 56 \\ - 22 \\ \hline 34 \end{array}$$

$$\begin{array}{r} 52 \\ \times 2 \\ \hline 104 \end{array}$$

* Vector

$$25 \div 5 = 5$$
$$\begin{array}{r} 5 \\ 5 \overline{) 25} \\ \underline{25} \\ 0 \end{array}$$

$$\frac{dy}{dx}$$

$$\int y dx$$

- Fraction

- Power 2^3

- Angle 

- Trigonometry

- Graph

- Vector → chapter

- differentiation

- Integration

Topic - ①

Fraction

→ divide

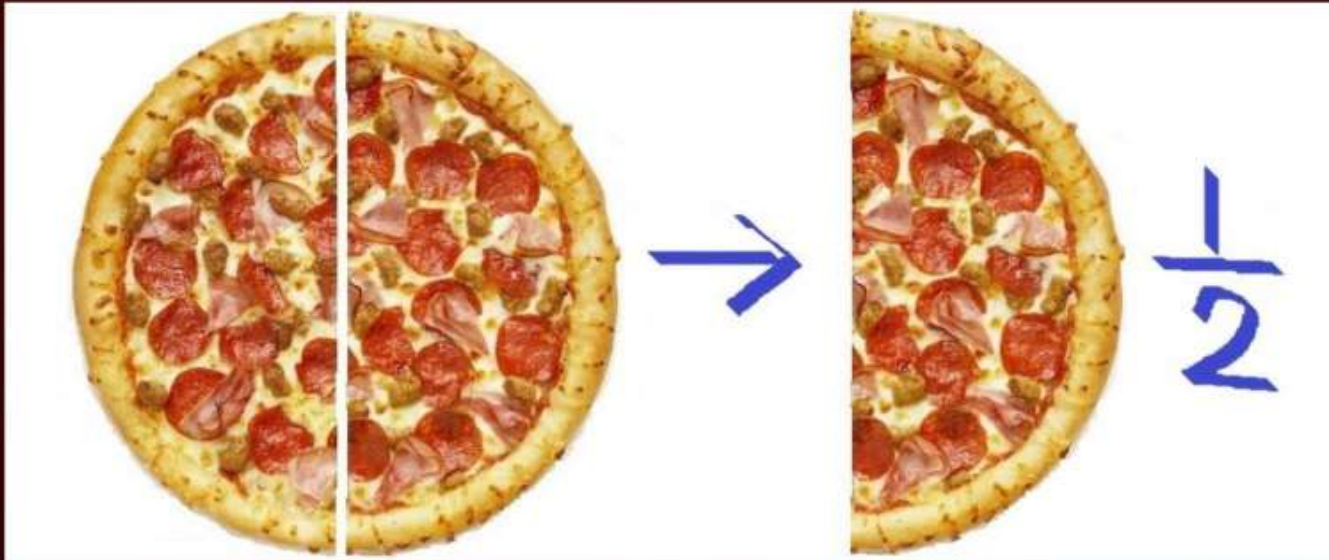
$$\frac{1}{2} = \underline{\underline{0.5}}$$

$$\underline{\underline{0}} \quad \underline{2 \times 0.5}$$

$$\underline{\underline{\text{Sol}}} \quad \cancel{2} \times \frac{1}{\cancel{2}}$$

$$= 1$$

Rule: 1



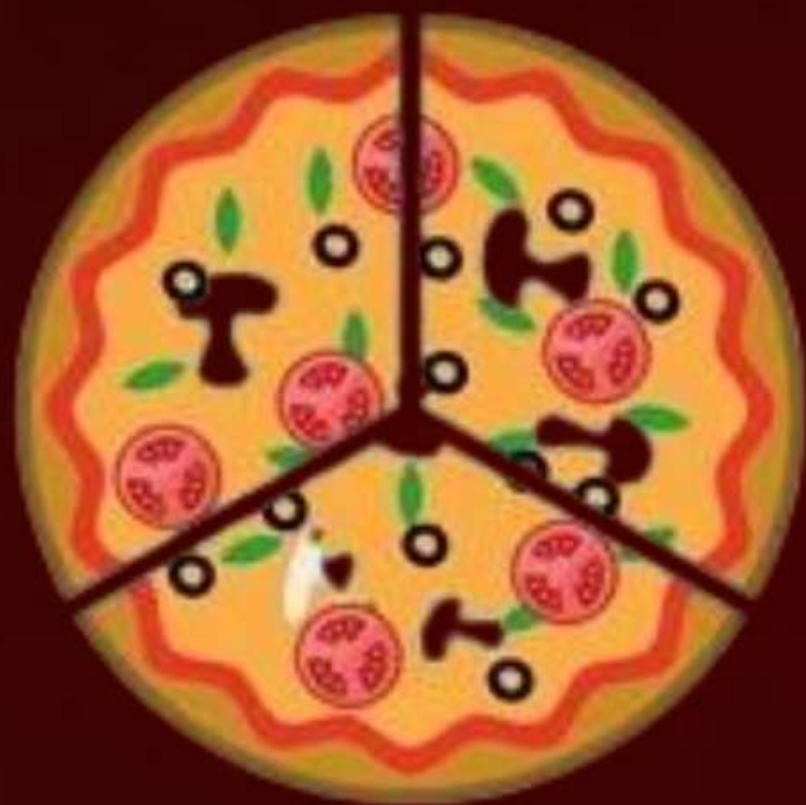
Rule 2

$$\frac{1}{3} = \underline{0.33}$$

$$\underline{Q} \quad 2 \times 0.33 = 0.67$$

$$2 \times \frac{1}{3} = 0.67$$

$$\frac{2}{3} = 0.67$$



$$\underline{Q} \quad 3 \times \underline{0.33} = 0.99$$

$$\cancel{3} \times \frac{1}{\cancel{3}} = 1$$

Rule 3

$$\frac{1}{4} = 0.25$$



$$\underline{\underline{Q}} \quad 2 \times 0.25$$

$$\cancel{2} \times \frac{1}{\cancel{4}_2}$$

$$\frac{1}{2} = 0.5$$

$$\underline{\underline{Q}} \quad 3 \times 0.25 = 0.75$$

$$3 \times \frac{1}{4} = 0.75$$

$$\frac{3}{4} = 0.75$$

$$\underline{\underline{Q}} \quad 4 \times 0.25 = 1$$

$$\cancel{4} \times \frac{1}{\cancel{4}} = 1$$



QUESTIONS 1

Q 0.33×9

$$\frac{1}{\cancel{3}} \times \cancel{9}^3 = 3 \checkmark$$

QUESTIONS 2

$$0.75 \times \frac{4}{3}$$

$$\underline{0.75} \times \frac{4}{3}$$

$$\frac{\cancel{1}3}{4} \times \frac{\cancel{4}^1}{\cancel{3}_1}$$

$$1 \times 1 = 1$$

rule 4

$$\frac{3}{10} = 0.3$$

$$\frac{2}{10} = \underline{0.2} \checkmark$$

$$\frac{3}{10} = 3 \times 10^{-1} = \underline{0.3}$$

$$\frac{4}{10} = 0.4$$

$$\frac{6}{10} = 0.6$$

$$\frac{8}{10} = 0.8$$

$$\frac{5}{10} = 0.5$$

$$\frac{7}{10} = 0.7$$

$$\frac{9}{10} = 0.9$$

rule 5



$$\frac{3}{100} =$$

$$\underline{\underline{Q}} \quad \frac{3}{100} = \frac{3}{10^2} = 3 \times 10^{-2} = \underline{\underline{0.03}}$$

$$\frac{4}{\underline{100}} = 0.04\underline{\underline{1}}$$

$$\frac{6}{100} = 0.06$$

$$\frac{8}{100} = 0.08$$

$$\frac{2}{100} = \underline{\underline{0.02}}$$

$$\frac{5}{\underline{100}} = 0.05\underline{\underline{1}}$$

$$\frac{7}{100} = 0.07$$

$$\frac{9}{100} = 0.09$$

QUESTIONS 1

$$\checkmark 678 \times 10^{-2} =$$

$$Q \ 5728 \times 10^{-2}$$

$$\frac{5728}{10^2}$$

$$678 \times 10^{-2}$$

$$\frac{5728.}{100}$$

$$\frac{678}{10^2} = \frac{678.}{100} = 6.\underline{7}\underline{8}$$

$$= \underline{57.28} \checkmark$$

rule 6

$$\underline{\underline{Q}} \quad \frac{49}{1000} =$$

$$\frac{49}{1000} = \frac{49}{10^3} = 49 \times 10^{-3} = \underline{\underline{0.049}}$$

$$\underline{\underline{Q}} \quad \frac{69}{1000} = \frac{69}{10^3} = 69 \times 10^{-3} = \underline{\underline{0.069}}$$

$$\underline{\underline{0}} \quad \frac{5}{10} = 0.5$$

$$\underline{\underline{0}} \quad \frac{5}{100} = 0.0\underline{5}$$

$$\underline{\underline{0}} \quad \frac{5}{1000} = 0.005$$

$$0 \quad \frac{5}{10000} = 0.000\underline{5}$$

rule 7

$$3.\overrightarrow{34} \times \underline{100} =$$

$$3.\overrightarrow{34} \times \underline{100} = 334.$$

$$6.64 \times 100 = 664$$

$$5.72 \times 100 = 572$$

$$\overset{0}{/} 62.54 \times 100$$

$$= 6254.$$

Rule ⑧

$$\rightarrow 64.2 \times 10 = 642$$

$$\rightarrow 64.232 \overset{\text{Right}}{\times 1000} = 64232.$$

Power

Topic: ②

• when you multiple the same number

$$2 \times 2 = 2^2 \xrightarrow{\text{power}} = 4$$

$$2 \times 2 \times 2 = 2^3 = 8$$

$$2 \times 2 \times 2 \times 2 = 2^4 = 16$$

$$2 \times 2 \times 2 \times 2 \times 2 = 2^5 = 32$$

$$W \times W = W^2$$

$$W \times W \times W = W^3$$



2

rule 1

$$\underline{10^0} = 1$$

$$\underline{9^0} = 1$$

$$\underline{100^0} = 1$$

$$\underline{a^0} = 1$$

$$x^0 = 1$$

$$w^0 = 1$$

$$\text{length}^0 = 1$$

Proof $x^0 = 1$

$$\frac{x^{1-1}}{x^1 \cdot x^{-1}}$$

$$x^1 \cdot x^{-1}$$

$$x^1 \cdot x^{-1} = x^{1-1}$$

$$\frac{x^1}{x^1}$$

$$\frac{x}{x} = 1$$

Proof

$$\begin{array}{l} x^0 = 1 \\ \rightarrow \underline{x^{1-1}} \end{array}$$

$$1 + -1 = 1 - 1$$

$$\underline{x^1} \underline{x^{-1}}$$

$$\frac{x^+}{x^+} = 1$$

rule 2

$$\underline{3^2} \times \underline{3^3} =$$

$$x^a \cdot x^b = \underline{x^{a+b}}$$

$$3^5 = 3 \times 5 = 15$$

X



$$\underline{3^2} \times \underline{3^3} = 3^{2+3} = 3^5$$

$$= 3 \times 3 \times 3 \times 3 \times 3$$

$$= 343$$

QUESTIONS



$$2^2 \times 2^3 =$$

$$2^2 \times 2^3 = 2^{2+3}$$

$$= 2^5$$

$$= 2 \times 2 \times 2 \times 2 \times 2$$

$$= 32$$

rule 3

$$5 \times 10^2 + 2 \times 10^2$$

$$\xrightarrow{\quad} \frac{x^a \cdot x^b = x^{a+b}}{x}$$

→ Sol

$$5 \times \underline{10^2} + 2 \times \underline{10^2}$$

$$10^2 (5 + 2)$$

$$10^2 \times 7 = 100 \times 7 = \underline{\underline{700}}$$

rule 4

$$(2^3)^2$$

$$(x^a)^b = x^{a \times b}$$

$$\begin{aligned} \text{Sol} \quad (2^3)^2 &= 2^6 = \underbrace{2 \times 2 \times 2 \times 2 \times 2 \times 2}_{= 64} \checkmark \end{aligned}$$

Note :-

$$x^2 = a$$

$$x = a^{\frac{1}{2}}$$

$$x = \sqrt{a}$$

under
Square Root (a)

$$x^3 = a$$

$$x = a^{\frac{1}{3}}$$

$$x = \sqrt[3]{a}$$

Cube Root ✓

rule 5

$$\sqrt{4} =$$

$$\sqrt{16} =$$

$$\sqrt{2} = \underline{1.4}$$

$$\sqrt{3} = \underline{1.7}$$

$$\sqrt{2} = 1.4$$

$$\sqrt{3} = 1.7$$

$$\rightarrow \sqrt{4} = \sqrt{2^2} = (2^2)^{1/2} = 2^{\cancel{2} \times \frac{1}{\cancel{2}}} = 2^1 = 2$$

$$\rightarrow \sqrt{16} = \sqrt{4^2} = (4^2)^{1/2} = 4^{\cancel{2} \times \frac{1}{\cancel{2}}} = 4^1 = 4$$

$$\rightarrow \sqrt{2} = 1.4^4 \quad \sqrt{3} = 1.7$$

QUESTIONS



$$(\sqrt{4})^{3/2}$$



$$(\sqrt{4})^{3/2}$$

$$(\sqrt{2^2})^{3/2}$$

$$2^{3/2}$$
 done

rule 6



$\frac{1}{\infty} = 0$

$\frac{1}{0} = \infty$

↑ Bohat Base number ↓

$$\frac{1}{\infty} = 0$$

$$\frac{1}{\infty} = 0$$

$$\frac{1}{0} = \infty$$



$$\frac{1}{2} = 0.5$$

$$\frac{1}{100} = 0.01$$

$$\underline{\underline{Q}} \text{ 2lakh} \times \frac{1}{\infty}$$

$$\begin{array}{c} \uparrow 3 \\ \frac{1}{3} = 0.33 \end{array} \downarrow \quad \uparrow \uparrow \uparrow \frac{1}{\text{Bohat Bar No.}} = 0$$

$$\underline{\underline{Sol}} \text{ 2lakh} \times 0$$

$$\frac{1}{4} = 0.25$$

$$\frac{1}{\infty} = 0$$

$$\begin{array}{c} \uparrow 5 \\ \frac{1}{5} = 0.2 \end{array} \downarrow$$

$$\frac{1}{0} = \infty$$

$$\frac{1}{10} = 0.1$$

$$= 0 \checkmark$$

Angle θ

Topic - (3)

:- Ratio of Arc to Radius

$$\text{Angle} = \frac{\text{Arc}}{\text{Radius}}$$

$$\theta = \frac{\text{Arc}}{R}$$

$$\theta = \frac{\pi}{\pi} = 1$$

Angle

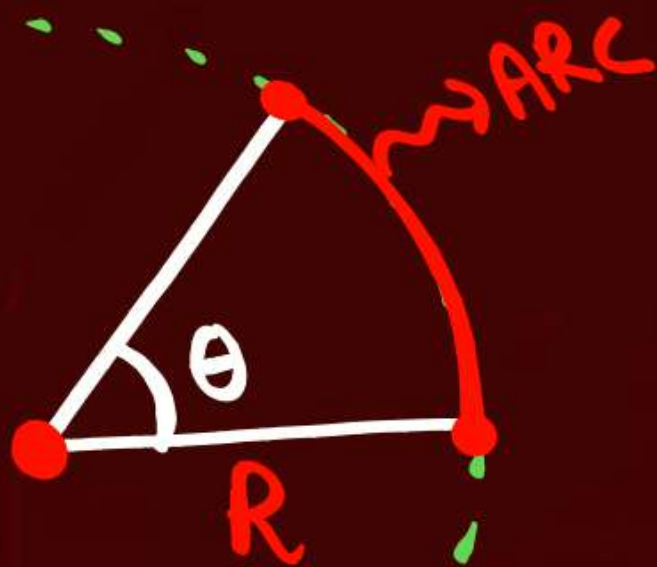
degree 45°

2 rad

Radian (S.I)

$$\# 1\text{m} = 100\text{cm}$$

$$180^\circ = \pi \text{ rad}$$



$$\frac{180^\circ}{180^\circ} = \frac{\pi \text{ rad}}{3.14 \text{ rad}}$$

$$\pi = 3.14 = \frac{22}{7}$$

$\pi = 3$

$$180^\circ = \pi \text{ rad}$$

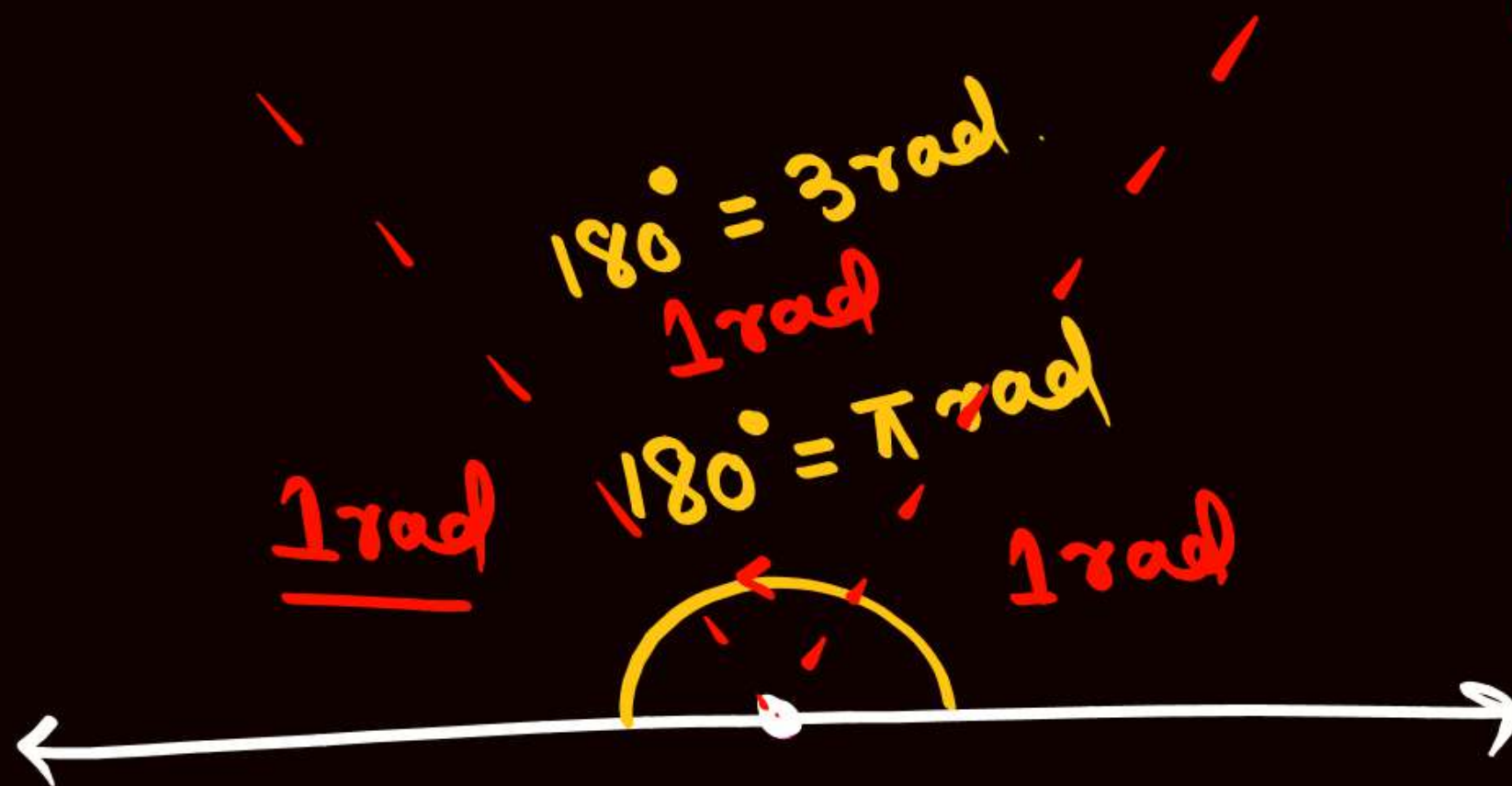
$$180 \times 1^\circ = \pi \text{ rad}$$

$$1^\circ = \frac{\pi}{180} \text{ rad}$$

$$\pi \text{ rad} = 180^\circ$$

$$\pi \times 1 \text{ rad} = 180^\circ$$

$$1 \text{ rad} = \frac{180^\circ}{\pi}$$



$$180^\circ = \underline{\pi \text{ rad}}$$

$$180^\circ = 3 \text{ rad}$$

degree $\longrightarrow$ Radian

Q

$$30^\circ = \underline{\hspace{1cm}} \text{ rad.}$$

Sol

$$30^\circ = 30 \times \underline{1^\circ} \quad 1^\circ = \frac{\pi}{180} \text{ rad}$$

$$= \cancel{30}^1 \times \frac{\pi}{\cancel{180}_6} \text{ rad}$$

$$= \frac{\pi}{6} \text{ rad.}$$

Q

$$60^\circ = \underline{\hspace{1cm}} \underline{\text{rad}}$$

$$= 60^\circ$$

$$= 60 \times 1^\circ$$

$$= \cancel{60}^1 \times \frac{\pi}{\cancel{180}_{360}} \text{ rad}$$

$$= \frac{\pi}{3} \text{ rad.}$$

$$Q \ 45^\circ = \underline{\hspace{2cm}} \text{ rad}$$

$$45^\circ$$

$$45 \times 1^\circ$$

$$1 \quad \cancel{45} \times \frac{\pi}{\cancel{4}180} \text{ rad}$$

$$\frac{\pi}{4} \text{ rad}$$

$$90^\circ = \underline{\hspace{2cm}} \text{ rad.}$$

$$90^\circ$$

$$90 \times 1^\circ$$

$$\cancel{90} \times \frac{\pi}{\cancel{180}2} \text{ rad}$$

$$\frac{\pi}{2} \text{ rad.}$$

$$\underline{\text{Q}} \quad 180^\circ = \pi \text{ rad}$$

$$180^\circ$$

$$180 \times 1^\circ$$

$$\cancel{180} \times \frac{\pi}{\cancel{180}} \text{ rad}$$

$$\pi \text{ rad}$$

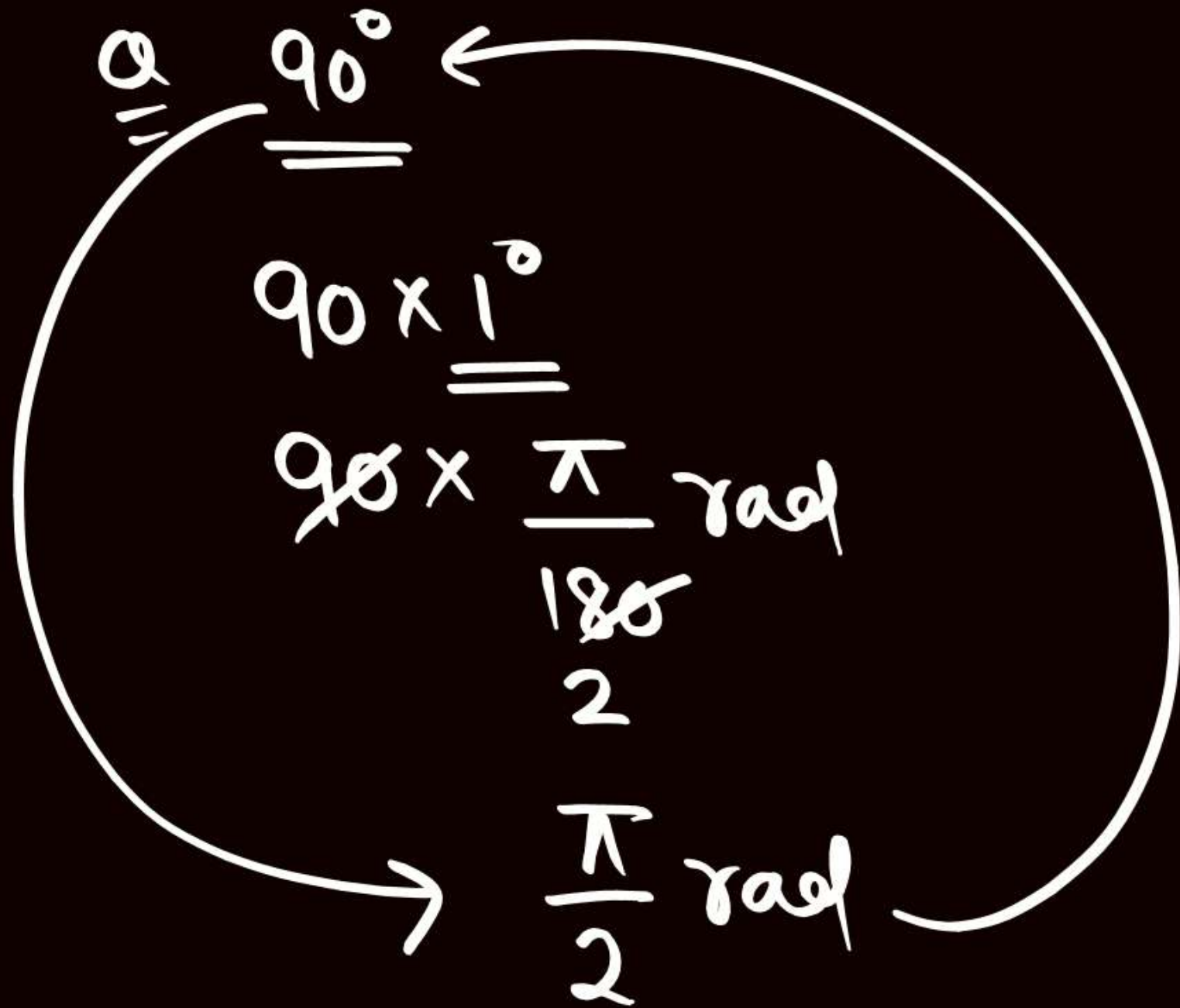
$$\text{Q } 360^\circ$$

$$360^\circ$$

$$360 \times 1^\circ$$

$$360^2 \times \frac{\pi}{\cancel{180}} \text{ rad}$$

$$\underline{2\pi \text{ rad}}$$



$$\underline{H/w} \quad \frac{\pi}{2} \text{ rad} = \underline{90^\circ}$$

Radian $\rightarrow$ degree

Q $\frac{\pi}{6}$ rad = _____°

$\frac{\pi}{6}$ rad

1 rad = $\frac{180^\circ}{\pi}$

$\frac{\pi}{6} \times (1 \text{ rad})$

$\frac{\pi}{6} \times \frac{180^\circ}{\pi} = \frac{180^\circ}{6} = 30^\circ \checkmark$

$$Q \quad \frac{\pi}{3} \text{ rad} = \underline{\hspace{2cm}}^\circ$$

$$\frac{\pi}{3} \text{ rad}$$

$$\frac{\pi}{3} \times 1 \text{ rad}$$

$$\frac{\cancel{\pi}}{3} \times \frac{180^\circ}{\cancel{\pi}} = \frac{180^\circ}{3}$$

$$= \underline{\underline{60^\circ}}$$

$$Q \quad \frac{\pi}{4} \text{ rad} = \underline{\hspace{2cm}}^\circ$$

$$\frac{\pi}{4} \times 1 \text{ rad}$$

$$\frac{\cancel{\pi}}{4} \times \frac{180^\circ}{\cancel{\pi}} = \frac{180^\circ}{4}$$

$$= 45^\circ \checkmark$$

Q $\pi \text{ rad} = \underline{\hspace{2cm}}^\circ$

Sol

$$\pi \text{ rad}$$

$$\pi \times 1 \text{ rad}$$

$$\cancel{\pi} \times \frac{180^\circ}{\cancel{\pi}}$$

$$\underline{\underline{180^\circ}}$$

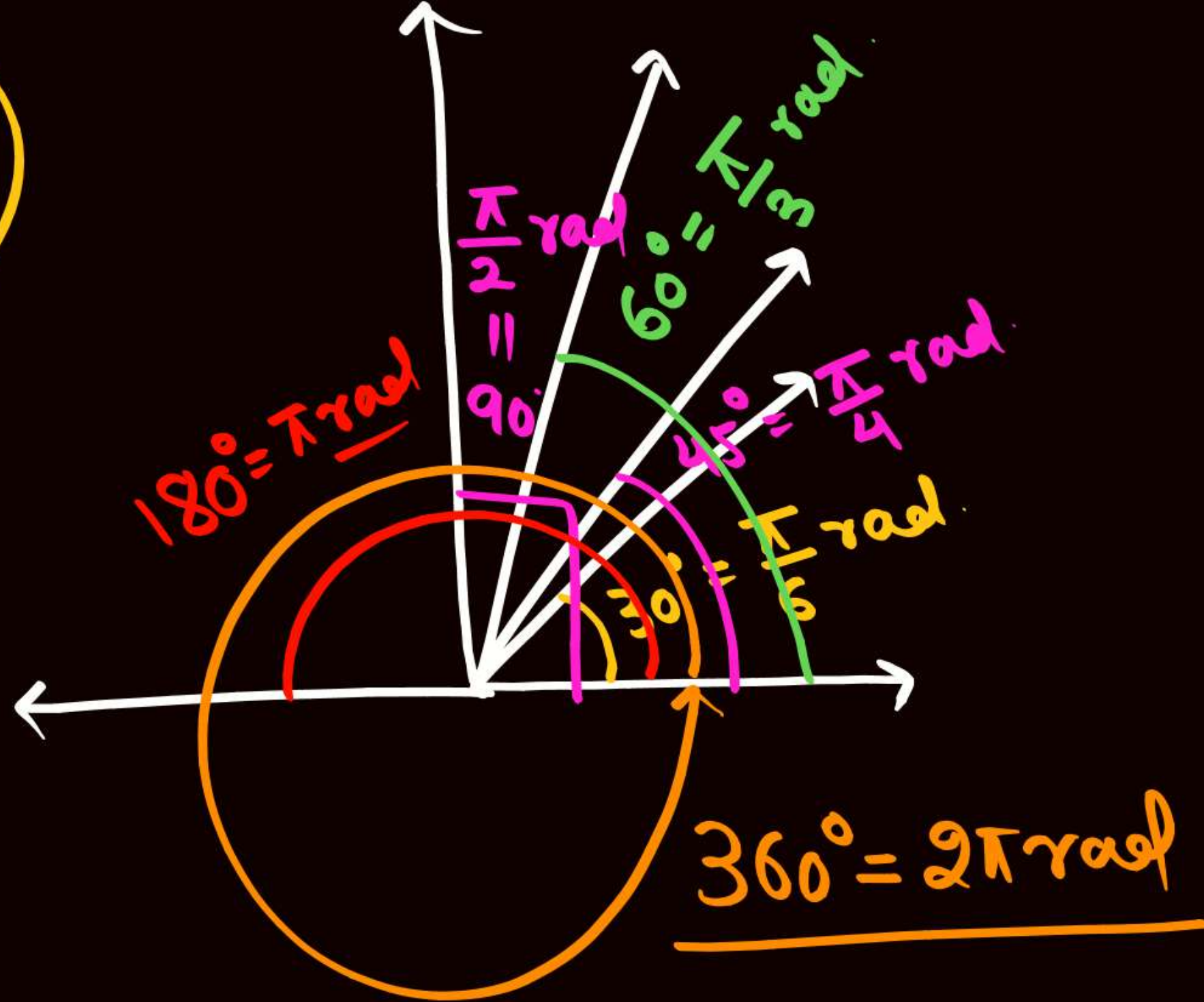
Q $2\pi \text{ rad}$

$$2\pi \times \underline{1 \text{ rad}}$$

$$2\cancel{\pi} \times \frac{180^\circ}{\cancel{\pi}}$$

$$2 \times 180^\circ \\ = 360^\circ$$

$$\frac{4}{\omega}$$



SUMMARY



<u>Fraction</u>	$\frac{1}{2}$	$\frac{2}{10}$	2.25×100
	$\frac{1}{3}$	$\frac{3}{100}$	$\frac{1}{\infty} = 0$
	$\frac{1}{4}$	$\frac{5}{1000}$	$\frac{1}{0} = \infty$

Power $\rightarrow 2^2$
 $2^0 = 1$
 $x^a x^b = x^{a+b}$

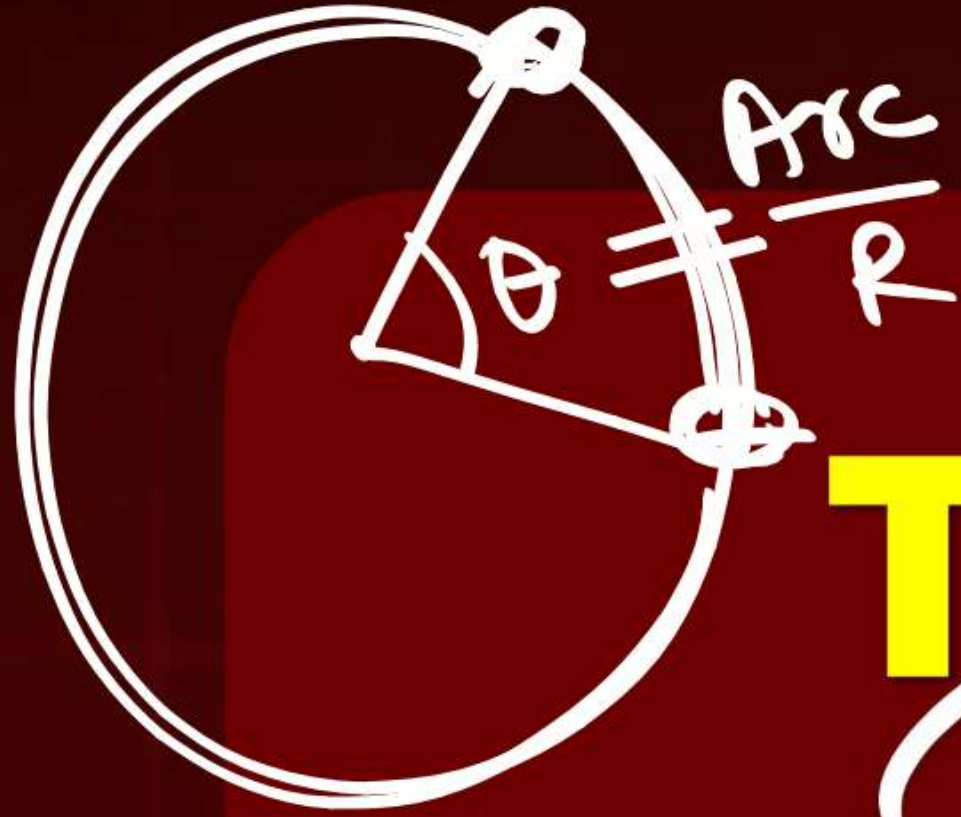
Angle $\rightarrow$ Rad $180^\circ = \underline{\pi \text{ rad}}$
 $\rightarrow$ degree

HOME WORK



$$5 \times 10^2 + 3 \times 10^2$$





Thank
You